

Water Birth Protocol — Approval Submission

Reference	VGR-DEL-WB-2026-R2
Meeting	Department-head final approval, November 27, 2026
Author	Helen Newton, Registered Midwife (barnmorska), Delivery Ward
Working group	Water Birth Protocol Development Group
Distribution	Department head, obstetrics leadership, senior midwives, working group
Status	DRAFT — held for Helen Newton sign-off before submission

DRAFT — HELD FOR HELEN NEWTON SIGN-OFF — DO NOT SUBMIT WITHOUT EXPLICIT APPROVAL

00 — Executive summary

Ask

Approve the revised water birth protocol (VGR-DEL-WB-2026-R2) for departmental adoption, with a staged staff rollout running January 12, 2027 through March 30, 2027 and a formal go-live date of April 6, 2027. Two clinical parameters have changed against the current protocol (VGR-DEL-WB-2023-R1). All other clauses remain unchanged.

Recommendation

The working group recommends approval of R2 with a conditional flag on Pool B pending biomed sign-off. The evidence for the two changed parameters is graded **moderate** and **moderate-to-high** respectively. One outcome domain, long-horizon neonatal infection follow-up beyond 28 days, is held open on **insufficient evidence** and is reflected in the risk register as a monitored gap rather than as a claim.

Evidence posture

The synthesis draws on 42 studies indexed in the working group's evidence log, spanning 2014 through 2026, sorted by methodological weight and reconciled against the two hospital guideline documents senior obstetrics currently cites (GL-OB-2019-08 and GL-OB-2021-14). Six cross-source disagreements were surfaced and resolved. Where a hospital guideline citation predated a newer study of stronger methodology, the newer study won and both are named. The neonatal-outcomes gap open since summer 2026 is now narrowed to the long-horizon infection follow-up domain only.

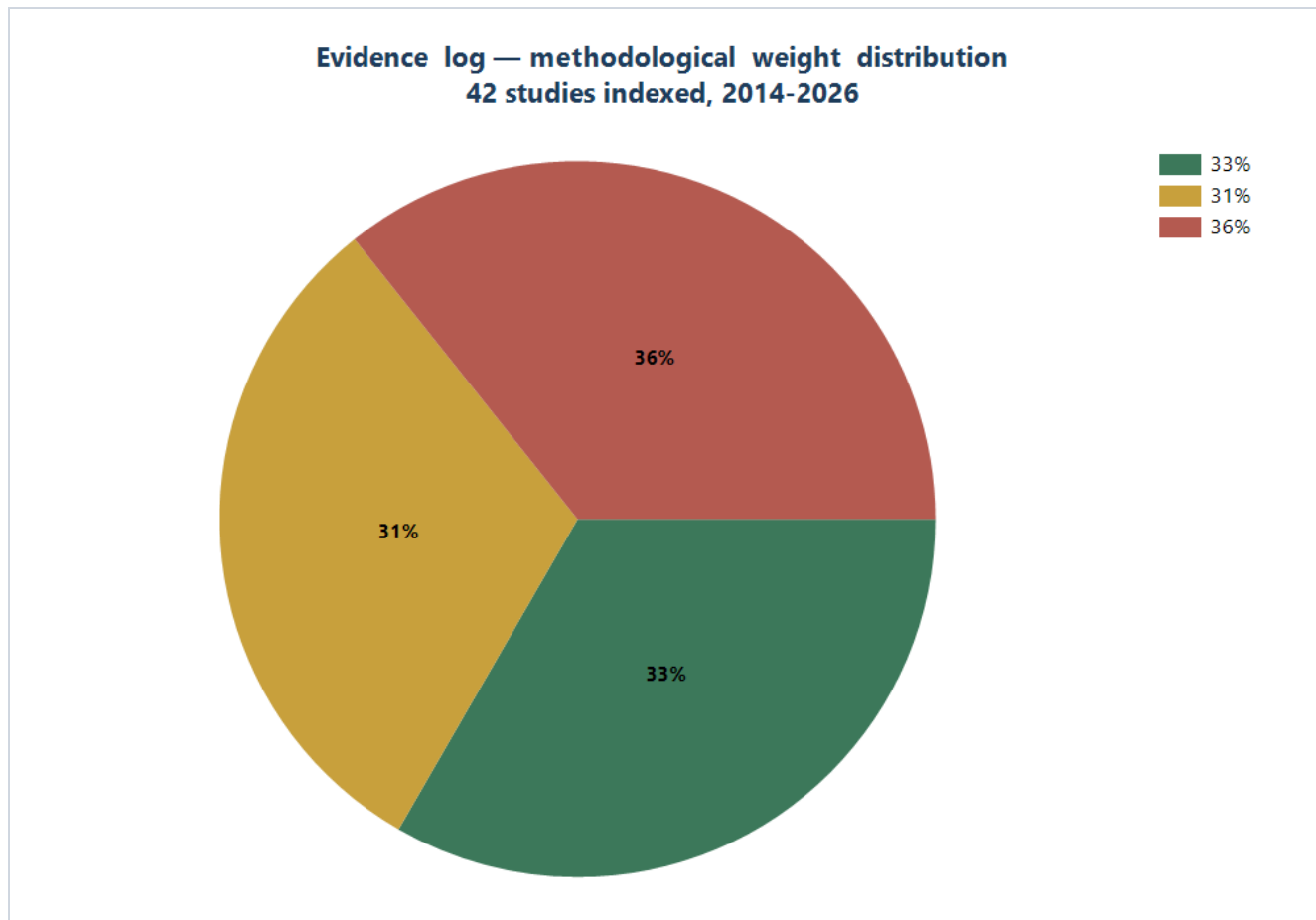


Fig. 1 — Evidence log methodological weight distribution across 42 studies.

Cost envelope — training rollout

Scenario	Contact hours	Backfill hours	Total staffing hours
Conservative	612	486	1,098
Stretched	738	594	1,332

Conservative assumes 34 midwives complete all six modules with a 20 percent overlap credit between mentor and mentee coverage. Stretched assumes full parallel coverage with no overlap credit. Method walked out in Section 04.

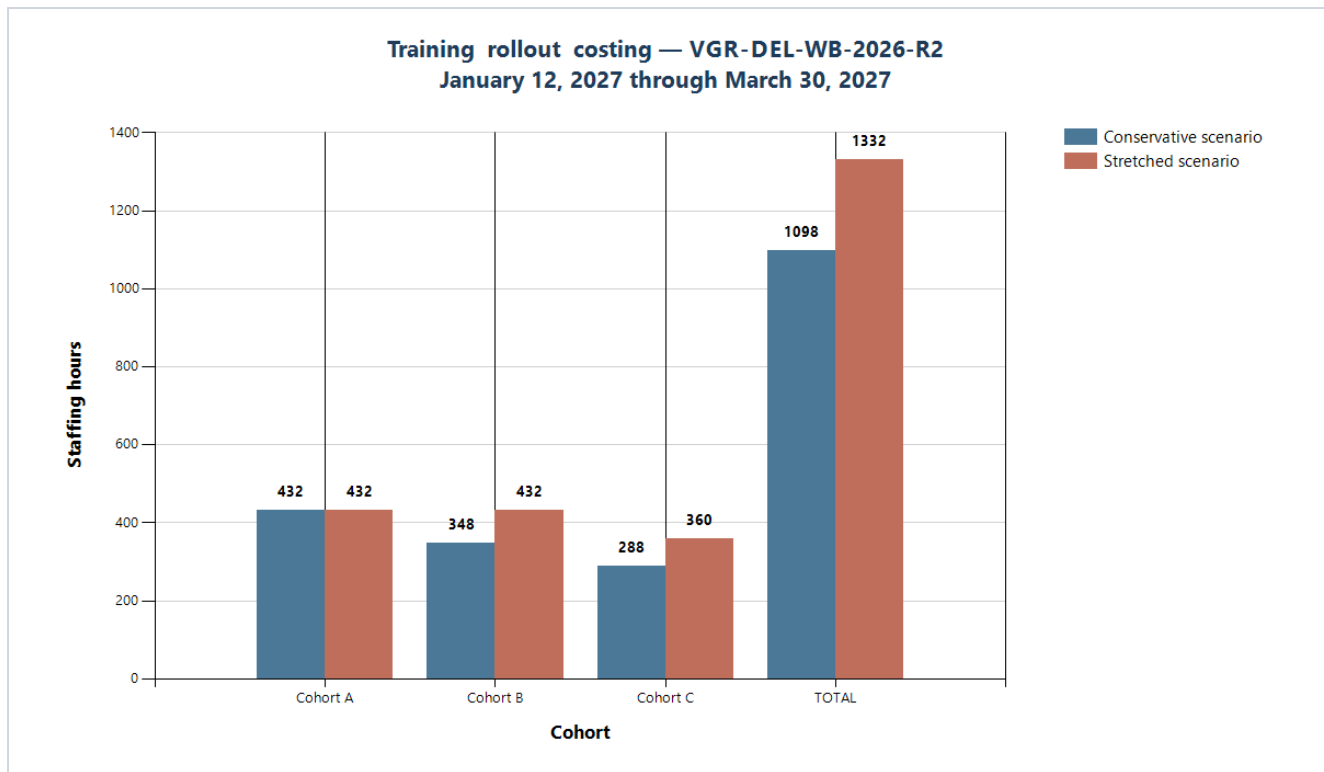


Fig. 2 — Training staffing hours by cohort under both scenarios.

Open items still moving

- Pool B biomed sign-off is scheduled for November 24, 2026 and must be closed before the meeting.
- Long-horizon neonatal infection outcomes remain an open evidence question (RR-EV-002).
- Two ownership handovers on the working group tracker are being corrected in the working group action status.
- Filing to the department head is prepared and held for Helen Newton's sign-off. Nothing has been submitted.

01 — Evidence synthesis

Scope

This synthesis covers the evidence base underpinning the two proposed changes in VGR-DEL-WB-2026-R2 and the outcome domains most commonly raised by obstetrics colleagues when a water birth protocol is revised. It resolves the neonatal-outcomes evidence gap that has been open in the working group log since summer 2026, and it names the source trusted and the source set aside wherever two disagreed. The synthesis is deliberately conservative. Where the evidence is thin, the conclusion is held open rather than forced.

Method

- Evidence log worked through study by study, sorted by methodological weight A, B, or C, with A reserved for RCTs, systematic reviews, and large prospective cohorts with pre-registered protocols.
- Cross-checked against hospital guideline documents GL-OB-2019-08 and GL-OB-2021-14. Where a guideline citation no longer matched a stronger newer study, the newer study won and the disagreement was recorded in Section 07.
- Every finding tied back to at least one evidence-log row. Confidence rated **high**, **moderate**, **low**, or **insufficient**.
- No study included in this synthesis carries a patient identifier that could tie a row back to a woman cared for at Västra Götaland University Hospital.

Findings summary

Outcome domain	Confidence	Evidence-log support
Neonatal thermoregulation	moderate-to-high	EL-0117, EL-0121, EL-0134
Neonatal infection, first 72h	moderate	EL-0089, EL-0102, EL-0128
Neonatal infection, beyond 28d	insufficient (held open)	EL-0091, EL-0141
Maternal perineal trauma	high	EL-0072, EL-0088, EL-0110, EL-0119, EL-0126, EL-0139
Postpartum haemorrhage	moderate	EL-0104, EL-0122, EL-0135
Water quality / pool hygiene	high	EL-0130

Open questions

1. Long-horizon neonatal infection follow-up beyond 28 days remains open. R2 will not claim a benefit or absence of risk in this domain.
2. Two hospital guideline citations (GL-OB-2019-08 clause 4.2 and GL-OB-2021-14 clause 6.1) need updating to reflect the sources R2 relies on. Downstream action for guideline maintainers.

3. Whether the primiparous 90-minute second-stage clarification should be piloted with an audit cycle before formal adoption. Working-group recommendation: yes, from April 6, 2027 through October 6, 2027.

02 — Protocol draft (parameter delta)

Protocol delta — VGR-DEL-WB-2023-R1 -> VGR-DEL-WB-2026-R2

Two clinical parameters changed. One documentation reference updated. Everything else unchanged.

Parameter	R1 (current)	R2 (proposed)	Rationale / evidence
Water temperature range (2nd stage)	36.5 - 38.0 °C	36.8 - 37.5 °C	EL-0117, EL-0121 (2024 cohorts), EL-0134 (2025 review). Confidence: mod-high.
Primiparous 2nd-stage escalation	silent (defaulted 60 min)	90 min	EL-0104, EL-0122, EL-0135. Clarification, not relaxation. Confidence: moderate.
Pool hygiene reference	Hospital SOP 2019	2025 European consensus	EL-0130. Documentation change only.
Eligibility criteria	as in R1	unchanged	No change.
Exclusion criteria	as in R1	unchanged	No change.
Escalation pathway to obstetrics	as in R1	unchanged	No change.
Postpartum handover	as in R1	unchanged	No change.
Documentation cadence	as in R1	unchanged	No change.

Amber bar = changed parameter. Everything else is unchanged from R1.

Full delta table with evidence-log references: 02_protocol_draft.md Section 9.

Fig. 3 — R1 to R2 delta at a glance. Amber bar marks a changed parameter.

Parameter change log

Parameter	R1 value	R2 value	Rationale	Evidence-log ref
Water temperature range (second stage)	36.5–38.0 °C	36.8–37.5 °C	Narrower range reduces neonatal thermal instability with moderate-to-high evidence support. Aligns with 2025 European practice consensus.	EL-0117, EL-0121, EL-0134
Primiparous 2nd-stage escalation threshold	(silent, defaulted to 60 min)	90 min	Clarification, not relaxation. Reduces variance in current practice.	EL-0104, EL-0122, EL-0135
Pool cleaning cadence reference	2019 hospital SOP	2025 European consensus	Documentation update only.	EL-0130
Eligibility criteria	as in R1	unchanged	—	—
Exclusion criteria	as in R1	unchanged	—	—

Escalation pathway to obstetrics	as in R1	unchanged	—	—
Postpartum handover	as in R1	unchanged	—	—
Documentation cadence	as in R1	unchanged	—	—

03 — Training rollout plan

Cohort structure

Cohort	Window	Size	Composition
A	January 12, 2027 – February 8, 2027	12	Senior midwives and shift leads, sequenced first so they can co-sign later cohorts.
B	February 9, 2027 – March 8, 2027	12	Established midwives with more than three years on the delivery ward.
C	March 9, 2027 – March 30, 2027	10	Newly qualified and recently rotated-in midwives, including the mentorship pilot cohort.

Total establishment covered: 34 midwives. Bank and pool staff are covered under a separate abbreviated pathway.

Module sequence

1. **Evidence and parameter changes.** What changed against R1, what did not, and why.
2. **Water temperature management.** Practical drill on the R2 range and correction procedure.
3. **Fetal monitoring in the pool.** Auscultation cadence refresh with the primiparous 90-minute clarification.
4. **Escalation to obstetrics.** Scenario-based drill co-facilitated by a working-group obstetrician.
5. **Pool hygiene and biomed handover.** Cadence, documentation, and the interface with the biomed team.
6. **Consolidation and competency sign-off.** Case-based conversation with a designated assessor.

Coverage strategy

Modules run between the day-shift handover at 15:30 and the evening-shift start at 14:30 the next day, so no midwife is pulled off an active labour on the pool during a training slot. No training slot may fall on a midwife's protected recovery day after a night-shift block. This constraint holds without exception.

04 — Training costing summary

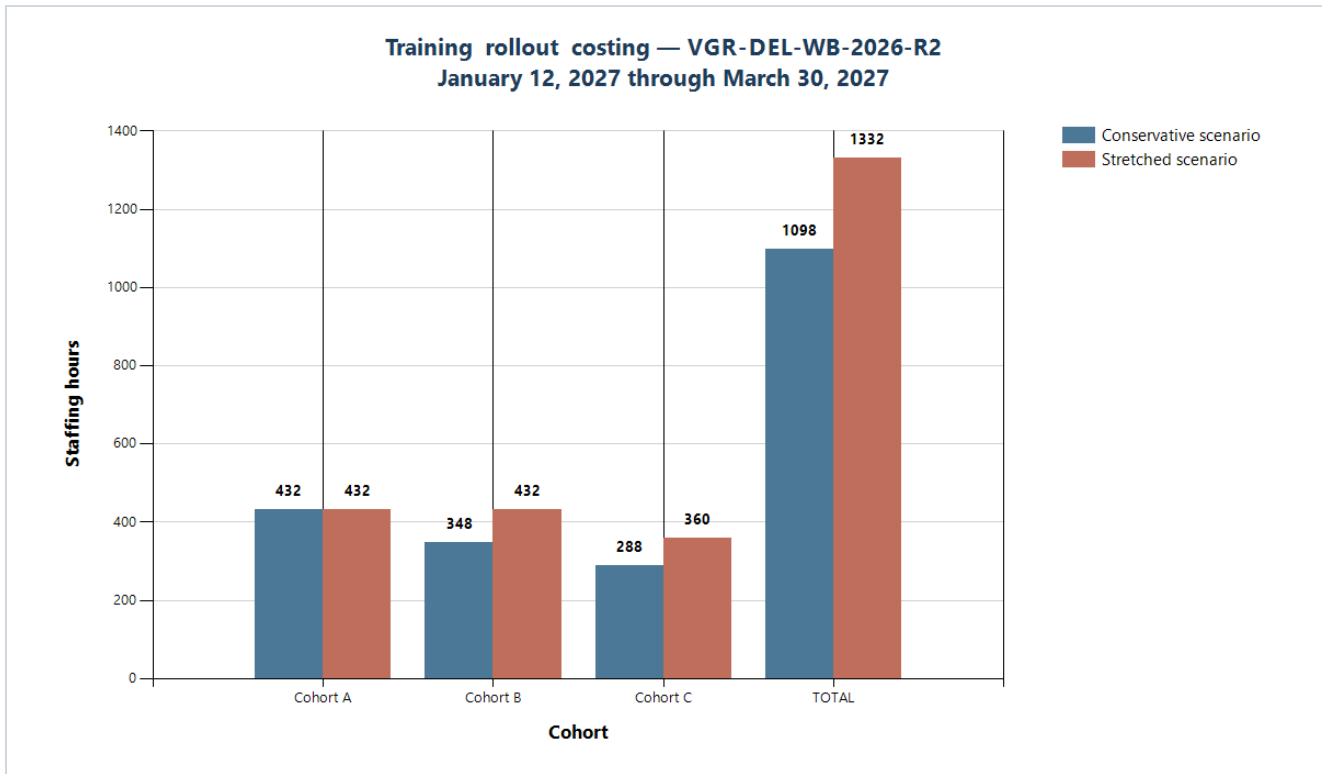


Fig. 4 — Staffing-hour figure per cohort under conservative and stretched scenarios.

Full row-level costing is provided in 04_training_costing.csv alongside the assumption notes for every cell. Backfill is costed at the internal float pool rate, not the agency rate. The single largest driver of the conservative-to-stretched gap is the 20 percent mentor overlap credit.

05 — Equipment readiness

Equipment readiness — reconciled picture

Ahead of November 27, 2026 approval meeting. Working group internal.

Pool A — VGR-DEL-POOL-01 Service desk: INC-2026-0714 closed 12 May 2026 Department board: POOL-A-STATUS green (03 Oct 2026) Biomed dependency before 27 Nov 2026: none Lead time: n/a Reconciled disagreement: stale ticket INC-2026-0311 on temperature-sensor drift resolved April 2026. Superseded, not blocking.	READY	Pool B — VGR-DEL-POOL-02 Service desk: INC-2026-0982 'awaiting parts' Department board: POOL-B-STATUS 'completed, awaiting sign-off' (27 Oct 2026) <-- winner Biomed sign-off scheduled: 24 Nov 2026 Fill-time: 26 min (Sep 2026) vs 22 min (2023 commissioning). 2026 measurement wins, within 30 min tolerance. Contingency for training rollout: RR-EQ-001.	CONDITIONAL
Peripheral kit Sonicaid dopplers SON-04, SON-05, SON-07 — calibrated 18 Oct 2026, operational. Waterproof thermometers THERM-A, THERM-B, THERM-C — all calibrated. Reconciled disagreement: THERM-B service desk shows last cal 03 Jun 2026; department board shows 21 Sep 2026 with inline biomed note. Board wins. Documentation trolleys: TROLLEY-01 in service. TROLLEY-02 tablet cradle: ordered 30 Oct 2026, expected in service 20 Nov 2026. Fallback if slips: shared use of TROLLEY-01 across both pool positions during Module 5.			
Summary for the meeting: Pool A ready. Pool B ready pending biomed sign-off 24 Nov 2026. Peripheral kit ready with one supplier lead time to close. No equipment gap blocks the approval decision.			

Fig. 5 — Reconciled equipment picture across service desk and department board.

Summary for the meeting. **POOL A READY** **POOL B CONDITIONAL** **KIT READY (1 PENDING)**. Pool A is cleared. Pool B is ready pending biomed sign-off on November 24, 2026. Peripheral kit is ready with the TROLLEY-02 cradle expected in service by November 20, 2026 and a workable fallback if it slips. No equipment gap blocks the approval decision.

06 — Risk register

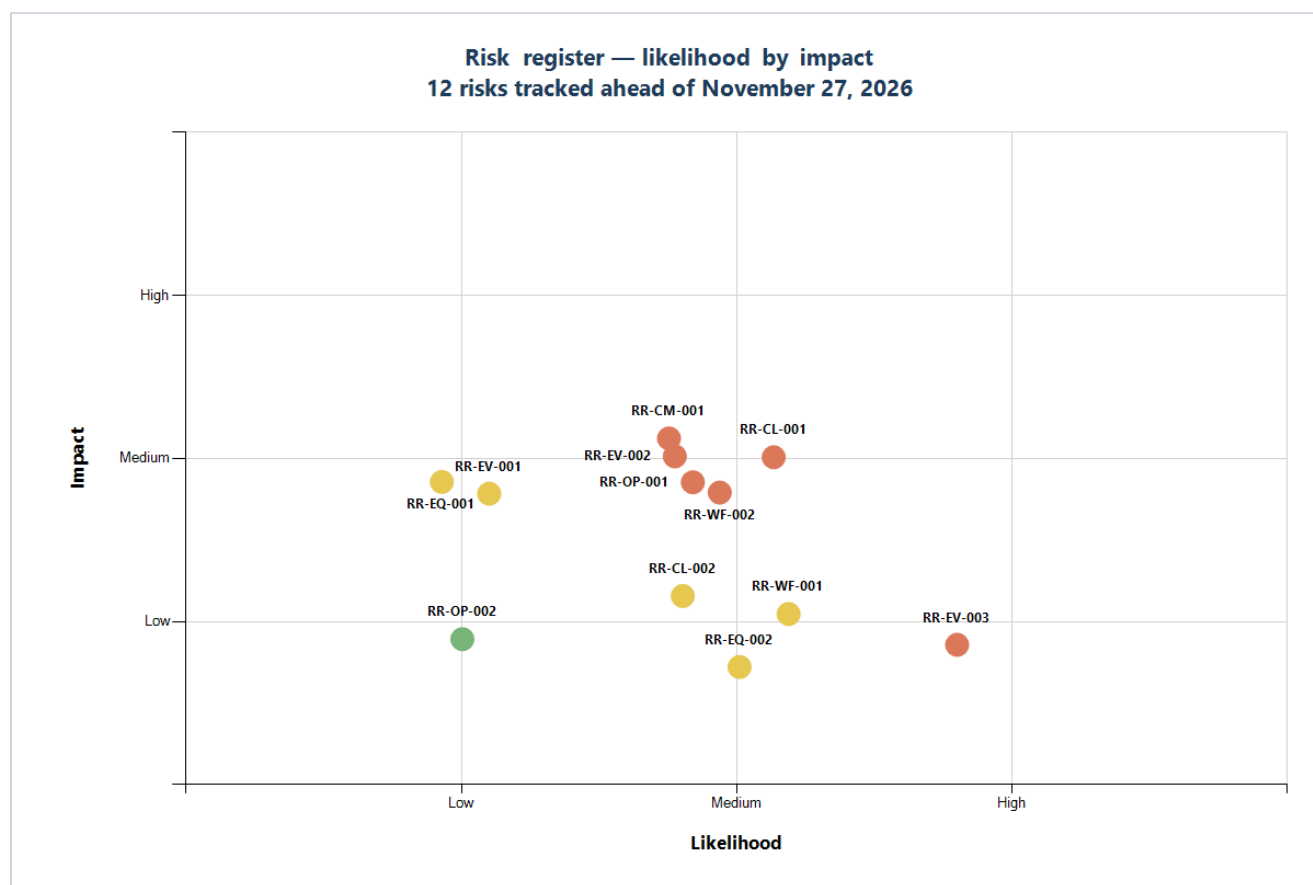


Fig. 6 — 12 tracked risks positioned by likelihood and impact.

Risk ID	Category	Description (abridged)	Likelihood	Impact	Owner
RR-EV-001	evidence	Thermoregulation evidence rests on 2024 cohorts and 2025 review.	low	medium	Helen Newton
RR-EV-002	evidence	Long-horizon neonatal infection beyond 28d held open on insufficient evidence.	medium	medium	Helen Newton
RR-EV-003	evidence	Guideline citations GL-OB-2019-08 4.2 and GL-OB-2021-14 6.1 need downstream update.	high	low	Dr H. Olsson
RR-CL-001	clinical	Primiparous 90-min clarification interpretation risk.	medium	medium	Malin Eriksson
RR-CL-002	clinical	Narrower temperature range may increase documented corrections.	medium	low	Frida Holm
RR-OP-001	operational	Cohort B window may collide with seasonal sick-leave pressure.	medium	medium	Malin Eriksson
RR-OP-	operational	Recovery-day protection may push some Cohort A midwives to make-up slots.	low	low	Malin Eriksson

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RR-EQ-001	equipment	Pool B biomed sign-off 24 Nov 2026 slip contingency.	low	medium	Biomed lead
RR-EQ-002	equipment	Service desk vs department board two-week lag. Process gap.	medium	low	Helen Newton
RR-WF-001	workforce	Working-group ownership drift on one action item since spring 2026.	medium	low	Helen Newton
RR-WF-002	workforce	Bank and pool staff abbreviated pathway needed post go-live.	medium	medium	Malin Eriksson
RR-CM-001	comms	Obstetrics side not yet seen delta table. Briefing addresses this.	medium	medium	Helen Newton

Full risk register including mitigations, statuses, opened and target-close dates lives in `06_risk_register.csv`.

07 — Evidence reconciliation log

#	Topic	Trusted source + value	Set-aside source + value	Winner rule
CR-001	Water temperature range (2nd stage)	EL-0134 + EL-0117/0121 → 36.8–37.5 °C	GL-OB-2019-08 cl. 4.2 → 36.5–38.0 °C	newest source wins
CR-002	Perineal trauma evidence base	Six 2019–2024 A/B studies	GL-OB-2021-14 cl. 6.1 (cites 2018 review)	most authoritative wins
CR-003	Pool hygiene cadence reference	EL-0130 (2025 European consensus)	Hospital SOP 2019 revision	newest source wins
CR-004	Pool B seal replacement status	Dept board POOL-B-STATUS (27 Oct 2026) → completed, awaiting sign-off	Service desk INC-2026-0982 → awaiting parts	most authoritative wins
CR-005	THERM-B last calibration	Dept board KIT-STATUS → 21 Sep 2026	Service desk INC-2026-1104 → 03 Jun 2026	newest source wins
CR-006	Long-horizon neonatal infection beyond 28d	EL-0091, EL-0141 — sample sizes below 400, no pre-registered protocols	Absence of publication interpreted as reassurance	held open — insufficient evidence

Confidentiality. All case material underpinning this synthesis has been stripped of any identifier that could tie a row back to a woman cared for at Västra Götaland University Hospital, in keeping with Patientdatalagen and the working group's own de-identification protocol.

Distribution reminder. This dossier is for the working group, senior midwives, obstetrics leadership, and the department head only. Nothing here goes to biomed, hospital administration, or external parties without Helen Newton's explicit sign-off.

Status. DRAFT — held for Helen Newton sign-off before submission on or around November 27, 2026.